

Assignment Instructions

Introduction

Thank you for applying. You will build a 15-screen conditional eligibility form for GLP-1 weight-loss medication screening. It must work smoothly end-to-end, with state management, persistence, branching logic, and a clean, accessible UI.

Time-box this to roughly 6-8 hours of focused work, ideally across 2-3 evenings. The goal isn't endurance — it's signal. Submit what you have at the end of your time-box; strong candidates use the write-up to explicitly call out what they cut and why. That's a feature, not a bug.

Stack

Please use these specific versions so we can review submissions on an even footing and so the patterns you write match our real codebase. If you can't use one for a strong reason, explain in the write-up.

- Next.js 15 (App Router), React 19, TypeScript 5.
- NestJS 11, Prisma 6, PostgreSQL 15.
- Vitest 4 for unit testing.
- Playwright Test 1.x (latest) for E2E.
- Tailwind 4 — or any modern alternative (CSS Modules, vanilla, etc.) is fine.
- Docker Compose for local DB.

Scope & Deliverables

Form Engine

- Use the logic defined in the Form Logic Specification tab.
- Represent it as a nested JSON schema containing: prompt text, input types, options for radio/checkbox, branching logic, and end states.

Frontend Application

- Build using React + Next.js (App Router).
- Each screen renders the prompt, the correct input type, and a Next button.
- Handle branch logic, state persistence, and resume flow on refresh.
- Responsive, smooth, accessible UI with a progress indicator.

Backend API

- Build using NestJS.
- Endpoints: start a new form session, save answers, return next question, return current state, return final evaluation result.
- Persist state in PostgreSQL with clean data models (sessions, answers).

State Management

- Frontend persists intermediate answers and saves to backend after each screen.
- Resume the current session on refresh.
- Handle concurrent users.

Documentation

- README with setup instructions, architecture decisions, and how to run tests.
- Brief design notes covering trade-offs and performance considerations.

Testing

For this role, testing is a first-class skill, not a final checklist item. The testing deliverables below carry significant weight in scoring.

Unit tests (Vitest 4)

- 100% branch coverage on the eligibility evaluator (the rules in Screen 15). The evaluator should be a pure function we can call from a test without spinning up Next.js or NestJS.
- API endpoint tests using `@nestjs/testing` with mocked Prisma.
- JSON-schema validation tests on the form-engine logic.

Playwright E2E

- Happy-path spec: a user clears Screens 1 → 15 with an Eligible result.
- Mid-flow refresh: open Screen 7, refresh the page, confirm state is restored and the user lands back on Screen 7 with prior answers intact.
- At least one terminal-state spec (Underage / Pregnancy / Already-on-GLP-1).
- At least one edge-case spec of your choosing — e.g. BP “Hypertensive Crisis” + “Normal” both checked. What does your UI do? Write the test that documents your decision.
- Use data-testid attributes so the tests don’t depend on visible-text selectors that break when copy changes.

CI workflow

- GitHub Actions workflow that runs Vitest + Playwright on PR open.
- Doesn’t need to be sophisticated — just demonstrates you can wire the gate.
- Bonus: cache node_modules and Playwright browsers between runs.

API Spec

- POST /api/session/start — start a new form session. Returns session ID + first question.
- POST /api/session/answer — save answer. Returns next question or final result.
- GET /api/session/:id — return saved progress.

AI Tools

Use AI coding tools (Claude Code, Cursor, GitHub Copilot, Claude API, etc.) as you would in your day job. We use them daily and expect you to as well — this is not a “prove you can do it without help” exercise. No tool restrictions.

What we want from you on this:

- Use the tools that make you most productive.
- Be honest about what you used in the write-up.
- Be ready to walk us through your tool choices in the pairing session (Stage 3): which tool you reached for when, what worked, what didn’t.

The signal we care about is your reasoning and how you compose the AI output into a coherent codebase, not whether you typed every keystroke.

Accessibility

This is patient-facing healthcare. Accessibility is a real requirement, not a checkbox.

- Form is keyboard-navigable end-to-end.
- Labels properly associated with inputs (<label htmlFor> etc.).
- WCAG 2.1 AA conformance for the inputs you build.
- Screen-reader-friendly error messages on validation failures.

Spec Ambiguities

The Form Logic Spec is real, but it has 2-3 unstated edge cases — situations where the spec doesn't say what should happen, or where the rule is ambiguous.

Strong candidates name these in their write-up and propose a resolution. Don't silently pick one; surface the question and how you decided. Healthcare software needs people who flag the unstated cases, not fill in blanks silently.

Reasoning Write-Up

A 1-2 page WRITEUP.md at the repo root covering:

- Three trade-offs you made and why.
- What you'd do differently with another week.
- Which AI tools you used and where; honest reflection on where they helped vs where they slowed you down.
- Any ambiguities you spotted in the Form Logic Spec and how you resolved them.

Evaluation Criteria

- Testing depth & rigour — 25%
- Correctness of eligibility logic — 25%

- Code quality / clean architecture — 20%
- Reasoning write-up — 15%
- API design — 10%
- UX / accessibility — 5%

What We Don't Want

- Live deployment URL — don't burn 2 hours on Vercel deployment; write tests instead.
- Load testing — irrelevant at this scope.
- Frontend animation polish — not what we're hiring for.
- Sub-millisecond performance tuning — not relevant.
- Custom design system from scratch — use Tailwind or any prebuilt; we want testing depth, not bespoke pixels.

If you have extra time, better directions are:

- Property-based tests for the eligibility evaluator (fast-check).
- Snapshot tests on the JSON-schema shape.
- An ADR (architecture decision record) for one non-obvious choice you made (e.g. why you picked Zustand over Context for state).
- A CI badge in the README.

What to Expect in Stage 3 (Pairing Review)

We will open your submission together and ask you to walk us through your reasoning.

Bring opinions about what you'd change. Reviewers will probe:

- Which AI tools you used and why.
- Why you made the testing choices you did.
- How you spotted (or missed) the spec ambiguities.
- How you'd scale your approach to the real Blume codebase.

Submission

Email a GitHub repo link and include WRITEUP.md at the repo root. We respond to every submission within 5 business days.

If you have any questions or doubts, feel free to reach out.

GLP-1 Eligibility Form Logic Specification

Overview

This document defines the conditional logic and content for a 15-screen self-screening form that determines eligibility for a GLP-1 weight-loss medication. Each screen is a separate interaction step. The form supports branching, multiple choice, radio buttons, and computed values. At the end of the flow, the system must evaluate all answers to determine one of:

- ✓ Eligible
- ✓ Ineligible
- ✓ Requires Clinical Review

The logic supports complex combinations and real-world medical screening criteria.

Screen 1 — Age

Prompt:

What is your age?

Type: Number

Branch Logic:

- If age < 18 → **End: Ineligible — Underage**
- If age > 75 → **Proceed with caution → Clinical Review Required**
- Else → Move to **Screen 2 — Weight**

Screen 2 — Weight (kg)

Prompt:

Enter your weight in kilograms.

Type: Number

→ Next: **Screen 3 — Height**

Screen 3 — Height (cm)

Prompt:

Enter your height in centimeters.

Type: Number

→ Next: **Screen 4 — BMI Evaluation**

Screen 4 — BMI Computation

Type: Computed

Compute: $BMI = \text{weight} \div (\text{height}/100)^2$

Branch Logic:

- $BMI < 25 \rightarrow$ **End: Ineligible — BMI Too Low**
 - $25 \leq BMI < 30 \rightarrow \rightarrow$ **Next: Screen 5**
 - $30 \leq BMI < 35 \rightarrow \rightarrow$ **Next: Screen 5**
 - $35 \leq BMI < 40 \rightarrow \rightarrow$ **Next: Screen 5**
 - $BMI \geq 40 \rightarrow$ **End: Requires Clinical Review — High BMI**
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Screen 5 — Pregnancy Status

Prompt:

Are you currently pregnant?

Type: Radio

Options: Yes / No

Branch Logic:

- Yes → **End: Ineligible — Pregnancy Contraindication**
 - No → **Next: Screen 6 — Comorbid Conditions**
-

Screen 6 — Comorbid Conditions (Multiple Choice)

Prompt:

Which chronic conditions have you been diagnosed with? (Select all that apply)

Type: Checkbox

Options:

- Hypertension
- Dyslipidemia
- Sleep Apnea
- GERD
- Thyroid Disorder

→ **Next: Screen 7 — Diabetes History**

(The count of selections influences final evaluation.)

Screen 7 — Diabetes History

Prompt:

Have you ever been diagnosed with diabetes?

Type: Radio

Options: Yes / No

Branch Logic:

- Yes → **Next: Screen 8 — HbA1c**
 - No → **Next: Screen 9 — Blood Pressure**
-

Screen 8 — Most Recent HbA1c (%)

Prompt:

Enter your latest HbA1c (%) result.

Type: Number

Branch Logic:

- $\text{HbA1c} > 9.0 \rightarrow$ **End: Ineligible — Uncontrolled Diabetes**
 - Else → **Next: Screen 9 — Blood Pressure**
-

Screen 9 — Blood Pressure Categories (Multiple Choice)

Prompt:

Check all that apply based on your most recent blood pressure reading

Type: Checkbox

Options:

- Normal (< 120/80)
- Elevated (120–129 / <80)

- Stage 1 Hypertension (130–139 / 80–89)
- Stage 2 Hypertension (≥ 140 / ≥ 90)
- Hypertensive Crisis (> 180 / > 120)

→ **Next: Screen 10 — Medications**

(Combination of selections influences final evaluation.)

Screen 10 — Current Medications (Multiple Choice)

Prompt:

Which medications are you currently prescribed?

Type: Checkbox

Options:

- ACE inhibitors
- Beta blockers
- Statins
- Thyroid medication
- GLP-1 receptor agonist

Branch Logic:

- If *GLP-1 receptor agonist* selected → **End: Requires Clinical Review — Already On Therapy**
 - Else → **Next: Screen 11 — Smoking Status**
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Screen 11 — Smoking Status

Prompt:

Do you currently smoke tobacco?

Type: Radio

- Yes
- No

→ Next: **Screen 12 — Alcohol Use Frequency**

Screen 12 — Alcohol Use Frequency

Prompt:

How often do you consume alcohol?

Type: Radio

- Never
- Monthly
- Weekly
- Daily

→ Next: **Screen 13 — Physical Activity Level**

Screen 13 — Physical Activity Level

Prompt:

How would you describe your typical activity level?

Type: Radio

- Sedentary
- Light (1–2x/week)
- Moderate (3–4x/week)

- Vigorous (5+x/week)

→ Next: **Screen 14 — Dietary Habits**

Screen 14 — Dietary Habits (Multiple Choice)

Prompt:

Which best describes your diet? (Select all that apply)

Type: Checkbox

- High sugar intake
- High processed foods
- Frequent sugary beverages
- High fiber diet
- Balanced diet

→ Next: **Screen 15 — Final Evaluation**

Screen 15 — Final Evaluation & Eligibility Outcome

No prompt: this screen performs evaluation based on all previous answers.

Possible Results

- ✓ Eligible
- ✓ Ineligible
- ✓ Requires Clinical Review

Rules Summary

Immediate Ineligibility:

- Age < 18

- BMI < 25
- Pregnancy = Yes
- HbA1c > 9.0
- GLP-1 receptor agonist selected

Automatic Clinical Review:

- Age > 75
- BMI ≥ 40
- Stage 2 + Diabetes present
- *Hypertensive Crisis* selected
- ≥ 3 comorbid conditions

Optional Review Rules:

- Stage 1 Hypertension + Sedentary + High sugar diet → Review
- Daily alcohol + moderate/high risk factors → Review

Example Decision Matrix

Path	Outcome
Age 45, BMI 32, No diabetes, BP normal	Eligible
Age 16	Ineligible
BMI 22	Ineligible

Pregnancy yes	Ineligible
HbA1c 10.2	Ineligible
BMI 42	Clinical Review
Hypertensive Crisis	Clinical Review